

Methodological Structure Definition Module

Module: MSDM — Methodological Structure Definition Module

Parent Standard: PMES — Prediction Methodology Standard

Architecture: PGA™ — Prediction Governance Architecture

Category: Governance & Enforcement

Subcategory: Prediction Governance

Governed Space: Methodological Structure Definition

Primary Governed Object: Prediction Methodology Structure

Module Type: Core Module

Canonical Language: UCL — Universal Canonical Language

Protection: MIP® — Methodological Intellectual Property

1. Canonical Definition

The Methodological Structure Definition Module (MSDM) governs the definition, organization, differentiation, relationship, sequencing, and structural coherence of the material components through which a selected predictive method operates as a Prediction Methodology.

MSDM establishes how the selected predictive method is structurally organized before its predictive logic, applicability, and methodology version are subsequently governed.

A selected method does not become a sufficiently governed methodology merely because its name, model, technology, or general methodological class is known.

Selected Predictive Method ≠ Defined Prediction Methodology Structure

2. Operational Role

MSDM is the second Core Module within PMES.

It receives the Selected Predictive Method from PMSM and establishes the structural organization through which that method participates in prediction construction.

Its principal transition is:

Selected Predictive Method → Defined Methodological Structure

MSDM governs what the methodology consists of and how its material components relate.

It does not yet provide the complete documentation of the predictive logic connecting those components to predictive output.

3. Internal Module Flow

Predictive Method Selection → → Methodological Structure Definition → Predictive Logic Documentation
→ Method Applicability → Methodology Version Governance

Current Position: Methodological Structure Definition

The defined structure proceeds into:



4. Operational Space

MSDM operates within the Methodological Structure Definition space.

It governs:

methodological component identification; component differentiation; component relationships; methodological sequencing; structural dependencies; methodological stages; human and machine roles; multi-method structure; structural completeness; structural coherence; structural change; structural traceability.

MSDM governs the architecture of the Prediction Methodology.

5. Operational Scope

Methodological structures may contain:

analytical procedures; statistical mechanisms; mathematical models; AI models; machine-learning systems; human judgment; expert review; aggregation mechanisms; transformation stages; simulation-derived components; market-derived mechanisms; decision rules; model ensembles; computational processes; other material predictive components.

MSDM does not prescribe a universal methodology structure.

It requires the actual structure used to remain sufficiently identifiable.

6. Methodological Component Identification

The material components forming the Prediction Methodology must be identifiable.

A component is material where its presence, absence, configuration, relationship, or function can materially affect how the methodology operates or what predictive output it can produce.

Not every technical implementation detail must become a separate governed component.

The governance requirement is material structural visibility.

Technical Detail ≠ Methodological Component by Default

7. Component Differentiation

Methodological components performing materially different functions must remain distinguishable.

A model, human reviewer, aggregation mechanism, simulation process, transformation stage, or market mechanism must not be represented as methodologically interchangeable where their functions differ.

Shared Methodology ≠ Same Methodological Function

This distinction becomes especially important in hybrid and multi-stage predictive systems.

8. Component Relationships

MSDM must establish how material methodological components relate to one another.

Relationships may include:

input relationship; dependency; sequence; transformation; aggregation; comparison; filtering; weighting; review; escalation; selection; combination; feedback; another structurally material relationship.

Identifying components without identifying their material relationships may leave the methodology structurally undefined.

Known Components ≠ Defined Methodological Structure

9. Methodological Sequencing

Where sequence materially affects predictive operation, MSDM must establish the order in which methodological components act.

For example:

Source Processing → Model Analysis → Expert Review → Aggregation

is methodologically different from:

Expert Judgment → Model Adjustment → Final Aggregation

even where the same components are present.

Same Components + Different Sequence ≠ Same Methodological Structure

10. Structural Stages

A methodology may operate through multiple stages.

MSDM must distinguish stages where they represent materially different methodological functions.

Stages may include:

preparation; transformation; analysis; modeling; comparison; aggregation; review; synthesis; output preparation.

Stage names themselves are not canonical requirements.

What matters is that materially distinct methodological functions remain structurally identifiable.

11. Structural Dependencies

A methodological component may depend upon another component, system, process, capability, or condition.

Material structural dependencies must remain identifiable where failure, replacement, or alteration of the dependency could materially change the methodology.

Component Presence ≠ Component Independence

MSDM governs structural dependency.

Prediction Basis source relationships remain governed by PBAS.

12. Mandatory and Conditional Components

A methodology may contain components that are:

Mandatory

Conditional

Optional

Alternative

Fallback

Where these distinctions materially affect how the methodology operates, they must remain identifiable.

A conditional component must not retrospectively be represented as though it had participated in every predictive instance.

Available Component ≠ Applied Component

13. Human Methodological Components

Human participation may form a structural component of Prediction Methodology.

Roles may include expert judgment, interpretation, review, adjustment, selection, escalation, or synthesis.

Where human intervention can materially alter the predictive process, its structural position must remain identifiable.

Human Involvement $\neq$ Undefined Methodology

Likewise:

Human Review $\neq$ Automatic Methodological Validation

14. AI and Machine Components

AI models, algorithms, machine-learning systems, and other computational mechanisms may form methodological components.

MSDM must establish their structural role rather than treating the presence of AI as a sufficient description of the methodology.

A methodology containing an AI model may also contain preprocessing, multiple models, retrieval systems, external tools, aggregation, human review, or other components.

AI Model Presence $\neq$ Defined AI Prediction Methodology

15. Multi-Model Structures

A Prediction Methodology may contain multiple models.

MSDM must establish their structural relationship where material.

Models may operate:

independently; sequentially; competitively; hierarchically; through voting; through weighting; through ensemble aggregation; through fallback logic; through another defined structure.

Multiple Models $\neq$ Ensemble by Default

16. Hybrid Methodological Structures

A methodology may combine human, computational, market-based, simulation-derived, statistical, or other predictive mechanisms.

MSDM governs how those mechanisms are structurally connected.

A hybrid methodology cannot be sufficiently described merely by listing its components.

Component List $\neq$ Hybrid Methodological Structure

The relationship among the components is part of the methodology.

17. Prediction Market Structures

Where prediction-market mechanisms form part of Prediction Methodology, MSDM identifies their structural position within the methodology.

A market-derived signal may function as:

primary predictive mechanism; one input into another method; comparison signal; aggregation component; adjustment mechanism; another defined methodological component.

MSDM does not determine whether market participants are independent, representative, information-advantaged, or concentrated.

Those governance questions belong principally to PBAS and CPGS.

18. Simulation-Supported Structures

Where simulation contributes to Prediction Methodology, MSDM establishes where simulation-derived outputs enter the predictive structure and how they relate to other methodological components.

Simulation itself remains governed by SIMULOS®.

Simulation Component ≠ Prediction Methodology in Its Entirety

The structural use of simulation within prediction does not transfer simulation governance to PGA™.

19. Structural Inputs and Outputs

Each material methodological component should have sufficiently identifiable relationships to what it receives and what it produces where those relationships are necessary to understand the methodology.

MSDM governs structural input-output relationships.

It does not determine the evidentiary quality or sufficiency of the information entering the methodology.

Methodological Input Position ≠ Prediction Basis Sufficiency

20. Structural Transformation

Where one methodological component transforms the output of another, the existence and role of that transformation must remain identifiable.

A transformation may materially affect the predictive meaning of downstream outputs even where the original source remains unchanged.

MSDM identifies the transformation stage.

PLDM subsequently governs the predictive logic explaining how that transformation contributes to prediction construction.

21. Structural Aggregation

Where multiple methodological components produce outputs that are subsequently combined, the aggregation point must remain structurally identifiable.

MSDM establishes where and between what components aggregation occurs .

It does not necessarily determine the detailed aggregation logic.

Structural Aggregation Point ≠ Aggregation Logic

Detailed predictive logic belongs to PLDM, while collective prediction governance may additionally belong to CPGS where multiple predictors are involved.

22. Structural Feedback

Some methodologies may contain feedback relationships in which outputs from one stage affect an earlier or parallel stage.

Where such feedback materially changes methodology operation, the feedback structure must remain identifiable.

A methodology with feedback is structurally different from an otherwise similar one-directional process.

Feedback Structure ≠ Linear Structure

23. Structural Completeness

A methodological structure is sufficiently complete when all components and relationships necessary to understand the material organization of the predictive method are represented.

Completeness does not require documentation of every software function, mathematical operation, internal parameter, or implementation detail.

Structural Completeness $\neq$ Exhaustive Technical Documentation

The appropriate level is the level necessary for methodological governance.

24. Structural Coherence

Components must form a coherent methodological structure.

A collection of individually legitimate methods does not automatically constitute a coherent Prediction Methodology.

MSDM must identify whether components can legitimately operate together within the selected predictive method.

Valid Components $\neq$ Coherent Methodology

Where material structural incompatibility exists, the methodology requires clarification or redesign before proceeding.

25. Structural Complexity

Methodologies may range from a single defined predictive mechanism to highly distributed human-machine systems.

MSDM does not reward complexity.

A structure should contain the components necessary to govern the methodology and no implied requirement exists to add methodological stages merely to create sophistication.

Structural Complexity $\neq$ Methodological Quality

26. Structural Change

A material change to methodological components, relationships, sequencing, dependencies, aggregation, or human-machine roles may constitute a methodological structure change.

Such change must remain identifiable.

MSDM establishes that the structure changed.

MVGM — Methodology Version Governance Module subsequently governs whether that change requires a new methodology version and how continuity is preserved.

Structural Change $\neq$ Same Methodology Structure by Default

27. Structural Definition Sufficiency

Methodological Structure is sufficiently defined when:

material components are identifiable; materially different functions are distinguishable; component relationships are identifiable; sequence is defined where material; dependencies are identifiable; mandatory and conditional components are distinguishable where relevant; multi-method relationships are defined; human and machine roles are visible where material; aggregation and feedback structures are identifiable where applicable; the overall structure is coherent; material structural gaps are not concealed.

Structural sufficiency does not establish Method Applicability or predictive correctness.

28. Governance Requirements

MSDM requires that:

1. material methodological components be identifiable; 2. components performing materially different functions remain distinguishable; 3. material component relationships be established; 4. methodological sequence be defined where relevant; 5. material structural dependencies remain identifiable; 6. mandatory, conditional, optional, and alternative components remain distinguishable where material; 7. human and machine roles remain structurally visible where they can materially affect prediction; 8. multi-model and hybrid structures define component relationships; 9. aggregation and feedback structures remain identifiable where applicable; 10. structural completeness be sufficient for methodological governance; 11. structural coherence be established; 12. material structural changes remain traceable.

29. Governance Logic

MSDM applies the following operational logic:

Selected Predictive Method

↓

Material Components Identified

↓

Component Functions Differentiated

↓

Component Relationships Established

↓

Sequence / Stages Established where applicable

↓

Dependencies Identified

↓

Conditional / Alternative Components Identified

↓

Multi-Method / Human-Machine Structure Established where applicable

↓

Aggregation / Feedback Structure Established where applicable

↓

Structural Completeness Assessment

↓

Structural Coherence Assessment

↓

Defined Methodological Structure

↓

PLDM — Predictive Logic Documentation Module

Where structural sufficiency is not established:

Structural Governance Gap → Clarification / Restructuring → Reassessment

30. Minimum Implementation Framework

A minimum implementation of MSDM must establish:

A. Selected Method Reference

The predictive method selected under PMSM.

B. Material Components

The components necessary to represent the methodology's material structure.

C. Component Functions

The methodological function performed by each material component.

D. Component Relationships

Material relationships connecting the components.

E. Methodological Sequence

The applicable order of operation where sequence matters.

F. Structural Dependencies

Dependencies capable of materially affecting methodology operation.

G. Component Status

Where applicable:

Mandatory / Conditional / Optional / Alternative / Fallback

H. Human–Machine Structure

The relationship between human and machine components where applicable.

I. Aggregation / Feedback Structure

Material aggregation or feedback relationships where present.

J. Structural Coherence State

Whether the components form a coherent Prediction Methodology structure.

K. Structure State

At minimum:

Undefined → Candidate → Clarification Required → Defined → Changed → Superseded

Only a sufficiently Defined methodological structure proceeds into predictive logic documentation.

31. Governance Outputs

MSDM may produce:

Defined Methodological Structure; methodological component map; component-function references; component relationships; methodological sequence; dependency structure; component-status references; human-machine structural relationship; aggregation structure; feedback structure; structural coherence state; methodology structure state; unresolved structural requirements.

The primary governed output is:

Defined Methodological Structure

This becomes the principal input into:

PLDM — Predictive Logic Documentation Module

32. Critical Distinctions

Selected Predictive Method $\neq$ Defined Prediction Methodology Structure

Technical Detail $\neq$ Methodological Component by Default

Known Components $\neq$ Defined Methodological Structure

Same Components + Different Sequence $\neq$ Same Methodological Structure

Component Presence $\neq$ Component Independence

Available Component $\neq$ Applied Component

Human Involvement $\neq$ Undefined Methodology

Human Review $\neq$ Automatic Methodological Validation

AI Model Presence $\neq$ Defined AI Prediction Methodology

Multiple Models $\neq$ Ensemble by Default

Component List $\neq$ Hybrid Methodological Structure

Simulation Component $\neq$ Prediction Methodology in Its Entirety

Methodological Input Position $\neq$ Prediction Basis Sufficiency

Structural Aggregation Point $\neq$ Aggregation Logic

Valid Components $\neq$ Coherent Methodology

Structural Completeness $\neq$ Exhaustive Technical Documentation

Structural Complexity $\neq$ Methodological Quality

33. Enterprise Application

MSDM allows organizations to move beyond labels such as “AI forecast,” “expert forecast,” “ensemble model,” or “market prediction” and establish what the predictive methodology actually consists of.

This is particularly important in complex predictive environments where a final prediction may pass through multiple models, external services, human interventions, transformations, simulations, and aggregation stages before becoming visible to a decision-maker.

Without structural governance, an organization may know which technologies are present while remaining unable to explain how they collectively form the Prediction Methodology.

34. Use Case 1 — AI Prediction Built from an Unknown Chain of Systems

An organization receives a high-impact prediction from an AI-based forecasting platform.

The final interface identifies the prediction as “AI-generated,” but the output actually passes through data transformation, retrieval systems, several specialized models, external analytical services, a simulation component, an aggregation layer, and human adjustment before publication.

The organization knows the final model name but does not possess a governed representation of how these components interact.

The governance question is:

Can an organization claim to know the methodology behind a prediction when it knows the technologies involved but cannot establish the structure through which they collectively produced the predictive result?

MSDM requires the material components, relationships, sequence, dependencies, and human-machine roles to become structurally identifiable.

Knowing the Components $\neq$ Knowing the Methodology Structure

35. Use Case 2 — Human Override Changes a Model Prediction

A predictive system produces a model-based probability estimate.

Before the prediction reaches decision-makers, a domain expert reviews the output and may increase, decrease, reject, or replace the model prediction based on professional judgment.

The published result continues to be described as a “model prediction.”

The governance question is:

Is the methodology still legitimately represented as model-based when an undocumented human intervention can materially determine the final predictive output?

MSDM requires the human intervention to be represented as part of the methodological structure where it can materially affect prediction formation.

The issue is not whether human intervention is good or bad.

The issue is whether the actual structure that produced the prediction is the structure being represented.

Presented Methodology ≠ Actual Methodology unless their material structures correspond.

36. Canonical Boundary

MSDM governs the identification, organization, differentiation, relationships, sequencing, dependencies, human-machine roles, aggregation structure, feedback structure, completeness, coherence, and change of the material components forming Prediction Methodology. It does not govern predictive method selection, detailed predictive logic documentation, final Method Applicability, methodology version governance, Prediction Basis sufficiency, assumptions, prediction formation, probability representation, feasibility, or predictive performance.

MSDM establishes what the Prediction Methodology consists of and how its material parts are structurally connected .

It does not yet establish the complete predictive logic by which those parts produce the prediction .

Canonical Closing Statement

A list of models, tools, experts, and systems does not define a methodology. Their governed relationships do.

MSDM makes the structure behind a prediction identifiable before its predictive logic can be legitimately documented and evaluated.

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